Convex Set

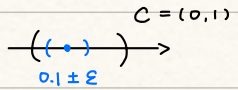
$\forall x_1, x_2 \in C$ 都有 $\alpha x_1 + \beta x_2 \in C$ $\quad C \subset \mathbb{R}$

$\alpha + \beta = 1$ $\quad \alpha + \beta = 1$ $\quad \alpha, \beta > 0$

$\alpha, \beta \in \mathbb{R}$ $\quad \alpha, \beta > 0$ 即可

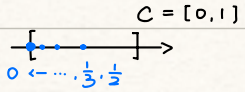
Topology

Open set：每个点都是内点



内点：存在开球在集合内 $\qquad$ 互为补集

Close set：每个聚点都属于 C

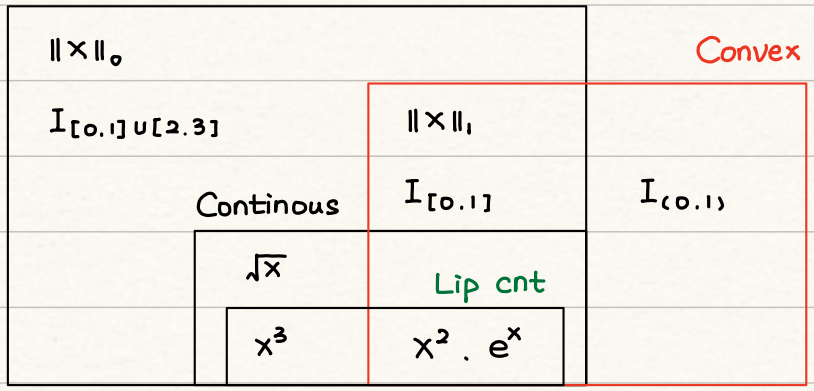


e.g. $\mathbb{R}, \Phi$ 既开又闭，$\{\frac{1}{n}\}_{n=1}^{\infty}$ 不开不闭

$[0, +\infty)$ 闭

Compact：Close + Bounded 在欧空中

Closed



Proper func：没有 $-\infty$，不全是 $+\infty$

Weierstrass：$f \in C^0([a,b])$，必有全局最优

S 紧集 or S 闭集且 f 强制

Coerveness 强制：$\lim_{x \to \infty} f(x) = \infty$ $\qquad$ e.g. $\inf e^x = 0$ 但 $\min e^x$ 不存在

Closed Function

Closed func 闭函数：Epi f 是闭集 $\overset{\not\Rightarrow}{\Leftarrow}$ 连续，dom f 闭 e.g. L_0 norm

e.g. C 是闭集，$I_C(x)$ 闭函数 $\Leftrightarrow$ Epi f 闭 $\Leftrightarrow$ $L_\alpha f$ 闭

$\Leftrightarrow$ LSC 下半连续

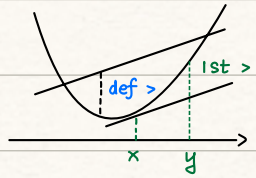
$$f(x) \le \lim_{n\to\infty} \min_x f(x_n)$$

Convex Function

Def. $\alpha f(x_1) + \beta f(x_2) \ge f(\alpha x_1 + \beta x_2)$ $\alpha+\beta=1$, $\alpha, \beta > 0$

$\le$ concave func

$=$ affine func affine $y = ax+b$ linear $y = ax$



Convex $\Leftrightarrow$ 1st $f(y) \ge f(x) + \langle \nabla f(x), y-x \rangle$

Convex $\Leftrightarrow$ 2nd Hess $f \ge 0$

Strictly convex $\alpha f(x_1) + \beta f(x_2) > f(\alpha x_1 + \beta x_2)$

$\Leftarrow$ Hess $f > 0$ e.g. x^4 strict, but $f''(0) = 0$

Prop. 凸 $\Rightarrow$ Lipschitz $\overset{\Rightarrow}{\not\Leftarrow}$ 连 e.g. $f(x) = \sqrt{x}$ 在 0^+ 连但不 Lipschitz

Lipschitz ctn，若 f 凸，x 内点 $|f(y) - f(x)| \le L\|y-x\|$

逐点上界：若 $f(x,y)$ 关于 y 凸，$g(y) = \min_x f(x,y)$ 必凸

$$g(\alpha_1 y_1 + \alpha_2 y_2) = \min_x f(x, \alpha_1 y_1 + \alpha_2 y_2)$$

$$\ge \min_x \alpha_1 f(x, y_1) + \alpha_2 f(x, y_2) \ge \alpha_1 \min_x f(x, y_1) + \alpha_2 \min_x f(x, y_2)$$

$$= \alpha_1 g(y_1) + \alpha_2 g(y_2)$$

Convex Problem

$Ax = b$

$\min f \quad \text{s.t. } h = 0 \,.\, g \le 0$

Def. f convex func, and $\{h = 0 \,.\, g \le 0\}$ convex

h linear, g convex func $\Rightarrow \{h = 0 \,.\, g \le 0\}$ convex

$\{h = 0 \,.\, g \le 0\}$ convex $\Rightarrow$ h 必线性